

Notes: Stroebe (JF, 2016)

Asymmetric Information about Collateral Values

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1 Big picture

- **Question.** What happens in credit markets when competing lenders are differentially informed about the quality of the collateral backing a loan?
- **Setting.** The paper studies newly built homes in Arizona from 2000 to 2011. In many housing developments, the property developer works with a vertically integrated mortgage lender that offers financing to buyers of the developer's homes.
- **Key source of information asymmetry.** Integrated lenders can observe or infer construction-quality information that is difficult for outside lenders and the econometrician to observe. This includes private information about the developer's homes, construction defects, foundation quality, and other latent collateral characteristics.
- **Main theoretical prediction.** An integrated lender uses its private signal to lend against better collateral. Nonintegrated lenders are adversely selected because they are more likely to finance homes rejected by the better-informed integrated lender.
- **Main performance result.** Homes financed by integrated lenders outperform observationally similar homes in the same developments financed by nonintegrated lenders by roughly 40 basis points per year in repeat-sales returns.
- **Default result.** Mortgages made by integrated lenders are less likely to enter foreclosure, with the relative foreclosure probability falling by roughly 40% in several specifications.
- **Collateral-not-borrower result.** The outperformance is not driven by observable borrower quality, maintenance, or second-owner characteristics. Integrated-lender houses continue to outperform during the second owner's holding period.
- **Private signal result.** Rejection by an integrated lender predicts lower future collateral returns and higher foreclosure probabilities. Rejection by nonintegrated lenders does not have the same predictive content.
- **Heterogeneity result.** Effects are stronger when private information about construction quality is more valuable, especially for houses built on expansive soil.
- **Pricing result.** Nonintegrated lenders charge higher mortgage interest rates when competing against integrated lenders, especially at higher loan-to-value ratios.
- **Takeaway.** Asymmetric information in mortgage markets is not only about borrower quality. Collateral quality itself can be privately known, affect lender competition, shape mortgage pricing, and predict foreclosure.

2 Incremental contribution of the paper

2.1 Contribution relative to borrower-information papers

- Much of the mortgage information literature focuses on asymmetric information about borrowers: income prospects, willingness to repay, or hidden borrower risk.
- Stroebe shifts attention to **collateral-value information**: lenders may know more about the quality of the house backing the mortgage.
- The paper shows that collateral information is empirically important even in a market where houses are appraised, inspected, and geographically proximate.

Interpretation: The incremental step is to treat the house, not just the borrower, as an object of private information in credit markets.

2.2 Contribution relative to collateral-channel research

- Many collateral-channel papers study how higher asset values relax borrowing constraints.
- This paper asks a different question: conditional on collateral being present, do lenders differ in what they know about its true quality?
- The answer is yes, and this information difference changes who lends, which houses perform well, and what interest rates uninformed lenders charge.

Interpretation: Collateral value is not just a public state variable. It can be privately known and selected into lenders' portfolios.

2.3 Contribution relative to empirical industrial organization of lending

- The institutional setting creates an unusually clear informed-versus-uninformed lender comparison.
- Vertically integrated lender-developer relationships give integrated lenders a plausible informational advantage about construction quality.
- The paper uses within-development comparisons, rejection data, soil heterogeneity, repeat sales, tax assessments, foreclosure records, and interest rates to test the mechanism.

Interpretation: The empirical design is stronger than a simple comparison of lender types because it uses multiple tests that point toward the same collateral-information channel.

2.4 Contribution relative to adverse selection and pricing theory

- The model predicts both quantity selection and price responses.
- Integrated lenders should lend against better collateral.
- Nonintegrated lenders should charge higher interest rates when competing against informed integrated lenders, especially when low collateral quality increases expected losses.
- The paper documents both effects, linking adverse selection to market pricing.

Interpretation: The paper connects information asymmetry to equilibrium mortgage pricing, not only to ex post collateral performance.

3 Data and measurement

3.1 Housing and mortgage data

- The main sample consists of newly built single-family residences in Arizona sold by developers from 2000 to 2007.
- The sample follows houses through subsequent sales, tax assessments, mortgage applications, and foreclosure records through 2011.
- The data allow the paper to identify whether the original mortgage was made by the developer's integrated lender or by a nonintegrated lender.
- Developments with an active integrated lender are the key empirical environment.

Interpretation: The setting lets the author compare houses built by the same developer in the same development but financed by different lenders.

3.2 Integrated lenders

- An integrated lender is a lender affiliated with, or cooperating closely with, the property developer.
- Integrated lenders have access to information from the developer that outside lenders may not observe.
- This information can include defects, foundation quality, construction quality, and other features not fully capitalized into sale prices at origination.

Interpretation: The integrated lender is not assumed to be intrinsically better at lending. It is better informed about the developer’s collateral.

3.3 Collateral performance outcomes

The paper measures ex post collateral performance over several horizons. Figure 1 defines the main windows.

- **Period A:** annualized return between the initial sale by the developer and the next arms-length resale.
- **Period B:** return from the initial sale to the January 2009 assessed market value.
- **Period C:** return during the second owner’s holding period.
- **Period D:** return between 2008 and 2009 assessed market values.
- **Foreclosure:** indicator for whether foreclosure is observed within three years of purchase.

Interpretation: Using multiple horizons helps distinguish true collateral quality from borrower selection, maintenance choices, price bundling, and market timing.

3.4 Borrower, financing, and housing controls

The main controls include:

- housing characteristics: initial price, lot size, building area, rooms, garage spaces, pool, owner occupancy, rental-unit status, and expansive soil;
- financing characteristics: loan-to-value ratio, loan-to-income ratio, mortgage duration, FHA or VA insurance, jumbo mortgage, and mortgage type;
- borrower characteristics: income, ethnicity, marital/household status, age, children, pet ownership, and census-tract demographics;
- fixed effects: month-of-sale, county, developer, development, and sales-quarter-pair fixed effects depending on the specification.

Interpretation: The paper’s key comparisons are often very local: houses in the same development, built by the same developer, sold in the same period, but financed by different lenders.

3.5 Auxiliary data

- HMDA mortgage application data identify whether a borrower was rejected by an integrated or nonintegrated lender.
- Tax assessment data provide assessed home values in January 2008 and January 2009.
- Building permit data identify substantial post-sale investment or maintenance.
- Soil data identify homes built on expansive soil, where construction quality is particularly important.
- Property-listing text identifies apparent damage or special-sale conditions.

- Interest-rate data for adjustable-rate and hybrid ARM mortgages identify price responses by nonintegrated lenders.

Interpretation: The auxiliary data allow the author to test alternative explanations one by one.

4 Model and empirical specifications

4.1 Adverse selection model

The model has two lender types. The integrated lender observes a signal about collateral quality; nonintegrated lenders observe only public information. Let the true collateral return be q_i , and let the integrated lender observe a signal

$$s_i = q_i + \eta_i. \quad (1)$$

The integrated lender conditions its financing offer on s_i . Nonintegrated lenders compete for borrowers but do not observe s_i .

Interpretation: Integrated lenders can select better houses into their own mortgage portfolios. Rejected houses then disproportionately flow to nonintegrated lenders.

4.2 Repeat-sales return specification

The main return specification is

$$\text{Return}_{i,q_1,q_2} = \alpha + \kappa \text{IL}_i + X_i' \beta + \delta_{q_1,q_2} + \varepsilon_i, \quad (2)$$

where:

- $\text{Return}_{i,q_1,q_2}$ is the annualized return of house i between sale quarter q_1 and resale quarter q_2 ;
- IL_i equals one if the initial mortgage was made by the integrated lender;
- X_i contains house, borrower, financing, and census-tract controls;
- δ_{q_1,q_2} are sales-quarter-pair fixed effects.

Interpretation: A positive κ means integrated lenders' collateral portfolios outperform nonintegrated lenders' portfolios after controlling for observables and market timing.

4.3 Tax-assessment return specification

The assessment specifications replace resale returns with returns from the initial sale to assessed values or from 2008 to 2009 assessments.

Interpretation: Tax assessments provide a performance measure even when houses do not transact. This addresses selection into resale.

4.4 Foreclosure specification

Foreclosure probability is estimated using probit specifications of the form

$$\text{Foreclosure3Years}_{i,m} = \alpha + \kappa \text{IL}_i + X_i' \beta + \delta_m + \varepsilon_i. \quad (3)$$

Interpretation: If integrated lenders know which homes have better collateral quality, their mortgages should be less likely to default through foreclosure.

4.5 Rejection-information specification

To test whether integrated lenders' private decisions are informative, the paper compares borrowers rejected by integrated versus nonintegrated lenders.

$$Y_i = \alpha + \kappa_1 \text{RejectedBy}IL_i + \kappa_2 \text{RejectedByNon}IL_i + X_i' \beta + FE + \varepsilon_i. \quad (4)$$

Interpretation: A rejection by the integrated lender should predict low future collateral returns if the integrated lender has private collateral information. Rejection by a nonintegrated lender should be less informative.

4.6 LTV heterogeneity

The paper estimates interactions with loan-to-value bins:

$$\text{Return}_i = \alpha + \sum_{j=1}^4 \kappa_j \text{LTV}_{ij} + \sum_{j=1}^4 \omega_j \text{LTV}_{ij} + X_i' \beta + FE + \varepsilon_i. \quad (5)$$

Interpretation: High-LTV borrowers are more sensitive to collateral value declines. The model predicts stronger adverse-selection pricing responses in high-LTV bins.

4.7 Mortgage interest rate specification

For ARM and hybrid ARM loans, the interest-rate regression is

$$\text{MortRate}_{i,m,f} = \alpha + \kappa \text{Has}IL_i + X_i' \beta + \delta_{m,f} + \tau_l + \varepsilon_i, \quad (6)$$

where $\text{Has}IL_i$ equals one if a nonintegrated lender competes against an integrated lender in the development-year.

Interpretation: If nonintegrated lenders anticipate adverse selection from informed competitors, they should charge higher rates in developments with integrated lenders.

5 Identification and empirical design

5.1 Main identifying comparison

- The main comparison is within developments that have an integrated lender.
- Houses financed by integrated lenders are compared to houses in the same developments financed by nonintegrated lenders.
- Developer, development, county, month, and sales-quarter fixed effects absorb common shocks and local market trends.

Interpretation: The key identifying variation is lender choice within very narrow housing markets.

5.2 Concern 1: borrower selection

- Integrated-lender borrowers look worse, not better, on many observable characteristics.
- The integrated-lender return advantage persists during the second owner's holding period, when the original borrower is no longer relevant.
- Rejections by integrated lenders predict poor collateral performance, while rejections by nonintegrated lenders do not.

Interpretation: These tests make it unlikely that the results are driven by unobserved borrower quality alone.

5.3 Concern 2: maintenance and investment

- Building permit data show no evidence that borrowers from integrated lenders invest more in maintenance or improvements.
- Assessment-based returns do not require resale and therefore reduce selection concerns.

Interpretation: The performance gap is unlikely to be driven by differential post-purchase maintenance.

5.4 Concern 3: unobserved house characteristics observed by all lenders

- If all lenders observed collateral quality equally, then nonintegrated lenders should price it through interest rates.
- Controlling for mortgage spreads does not eliminate the integrated-lender collateral advantage.
- Property listing data show integrated-lender homes are less likely to be listed as damaged or “as-is.”

Interpretation: The evidence points to private information held by integrated lenders, not just omitted public quality signals.

5.5 Concern 4: house-price discounts or bundled pricing

- If integrated lenders simply bundled mortgage discounts into initial house prices, later resale returns could mechanically look higher.
- The persistence of outperformance over tax-assessment horizons and second-owner horizons is hard to reconcile with pure initial price bundling.

Interpretation: The results are better interpreted as true collateral quality selection than as mechanical initial-price discounts.

6 Tables and figures

6.1 Table I and Figure 1: sample and performance horizons

Read:

- Table I reports summary statistics for house, borrower, financing, and census-tract controls.
- Borrowers from integrated lenders generally look worse on observables: lower income, smaller and cheaper houses, and more FHA-insured loans.
- Figure 1 defines four performance windows: repeat sale, tax assessment, second-owner period, and 2008–2009 assessment.

Economic message: Integrated-lender outperformance is unlikely to be driven by better observable borrowers. Figure 1 clarifies why multiple horizons are useful for identification.

Table I
Summary Statistics of Control Variables

This table shows mean, standard deviation, and median for the control variables in the analysis. The second-to-last column shows the coefficient on IL in the regression $Characteristic_i = \alpha + \beta IL_i + \psi year \times Development$ for developments with an integrated lender. The last column contains standard errors of the estimate, clustered at the development level and presented in parenthesis.

	No Integrated Lender			Has Integrated Lender			ΔIL	SE
	Mean	SD	Med.	Mean	SD	Med.		
Housing Characteristics								
Sales Price (k year-2000\$)	227.2	120.8	188.8	185.9	82.1	162.6	-3.19	(0.7)
Lot Size (Sq ft)	8,824	4,903	7,716	7,528	3,584	6,777	-56.6	(31.5)
Building Area (Sq ft)	2,362	862.8	2,185	2,117	720.2	1,966	-53.3	(7.82)
Price/Sqft (k year-2000\$)	29.4	13.3	26.2	27.5	9.7	25.5	-0.40	(0.17)
Garage Spaces	2.46	0.57	2	2.29	0.45	2	-0.01	(0.01)
Total Rooms	7.59	1.91	7	7.18	1.70	7	-0.14	(0.03)
Has Pool	0.39			0.29			0.01	(0.00)
Owner Occupied	0.80			0.80			0.06	(0.00)
Expansive Soil	0.06			0.10			-0.00	(0.00)
Financing Characteristics								
LTV Ratio	0.84	0.14	0.80	0.86	0.14	0.90	-0.00	(0.00)
LTI Ratio	2.49	1.01	2.41	2.58	1.05	2.50	0.19	(0.02)
Mortgage Duration (Years)	29.2	3.74	30	29.4	3.10	30	-0.22	(0.04)
FHA Insured	0.11			0.20			0.07	(0.01)
VA Insured	0.04			0.05			0.01	(0.04)
Jumbo Mortgages	0.01			0.00			-0.01	(0.00)
Borrower Characteristics								
Income (k year-2000\$)	88.8	71.7	71.8	73.4	58.6	60.2	-11.4	(1.10)
Single Person	0.33			0.36			-0.06	(0.01)
Latino	0.10			0.12			-0.02	(0.00)
Asian	0.04			0.04			-0.01	(0.00)
Single Parent				0.09			-0.00	(0.00)
Number of Children				0.56	0.85	0	0.03	(0.01)
Age of Borrower				47.3	17.3	46	3.16	(0.26)
Dog Owner				0.17			0.03	(0.00)
Cat Owner				0.09			0.02	(0.00)
Census Tract Demographics								
Med. Tract Inc. (k \$)	82.6	21.2	85.7	74.1	18.0	71.0		
High School Grad. Rate	0.91	0.10	0.95	0.89	0.09	0.92		

Table I: Summary statistics

6.2 Table II and Figure 2: repeat-sale outperformance and timing

Read:

- Table II shows that integrated-lender houses earn about 0.4 percentage points higher annualized repeat-sales returns.
- The effect remains stable after adding housing, borrower, financing, census-tract, developer, and development fixed effects.
- Figure 2 shows that the performance difference is driven by a thicker left tail among nonintegrated-lender properties.
- The bottom panel shows outperformance across different time gaps between sales.

Economic message: Integrated lenders select collateral with less downside risk. The result is not concentrated in one resale horizon.

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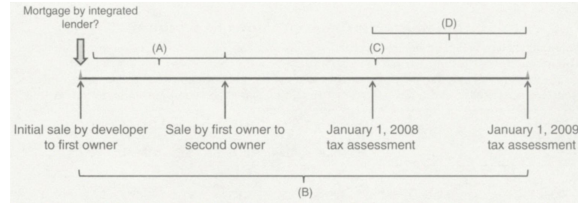


Figure 1. Measures of ex post price performance. This figure shows the four different time horizons over which ex post price performance is measured in the paper.

Figure 1: performance windows

Table II
Annualized Return (%): Between Repeat Sales, Period A

This table shows results from regression (1). The dependent variable is the annualized return of houses between two arms-length transactions. I include single-family residences first sold by a developer over the 2000 to 2007 period in developments with an integrated lender. In columns (6) and (7), I restrict the sample to those transaction pairs where the second transaction was preceded in the prior six months by a death or divorce of the initial owners. All specifications include sales quarter-pair fixed effects and county fixed effects. House characteristics (H) include real initial sales price, building size, lot size, price per square foot, number of garage spaces, average size of bedrooms and bathrooms, whether the house has a pool, whether the house is built on expansive soil, and whether the house is a rental unit. Buyer characteristics (B) include real income, number of children, age of the buyer, whether the property was purchased by an individual or a couple, whether the buyer is Asian or Latino, whether the buyer is a single parent, and whether the buyer is a dog or cat owner. Financing characteristics (F) include mortgage type, loan-to-value ratio, loan-to-income ratio, and mortgage duration. Census tract demographics (T) include median household income and the percentage of adults over 25 with a high school diploma. D1 includes developer fixed effects, D2 includes development fixed effects. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	FORCED MOVES	
						(6)	(7)
Integrated Lender	0.418*** (0.155)	0.455*** (0.148)	0.527*** (0.133)	0.420*** (0.115)	0.386*** (0.109)	0.409** (0.170)	0.365* (0.198)
Controls		H	H, B, F	H, B, F	H, B, F	H, B, F	H, B, F
				T, D1	T, D2		T, D1
R ²	0.869	0.877	0.878	0.887	0.896	0.885	0.903

Table II: repeat-sale returns

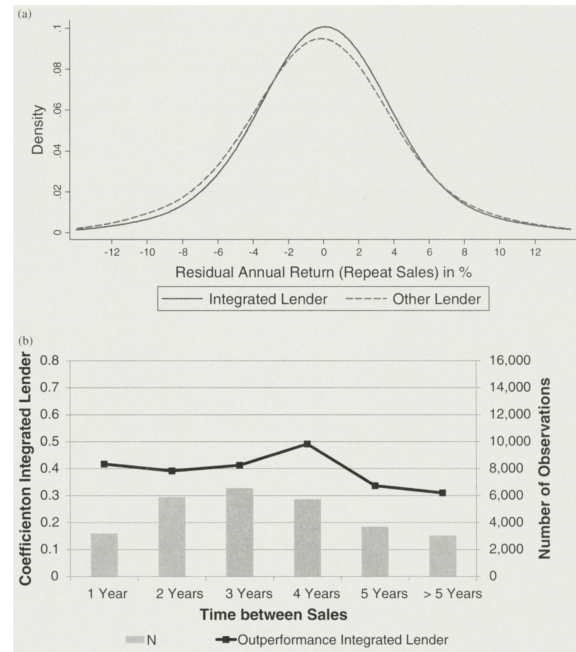


Figure 2. Density and timing of information release. The top panel plots the density of ϵ_i for
Figure 2: density and timing of information release

6.3 Tables III and IV: assessed values and foreclosures

Read:

- Table III uses tax-assessed market values instead of resale prices and finds similar integrated-lender outperformance.
- Table IV shows that integrated-lender mortgages are less likely to enter foreclosure within three years.
- The foreclosure coefficient is economically large relative to average foreclosure rates.

Economic message: The results are not driven by resale selection. Better collateral performance is also visible in assessed values and defaults.

Table III
Annualized Return (%): Initial Sale to Assessment, Period B

This table shows results from regression (3). The dependent variable is the annualized return of houses between the initial sale and the January 2009 assessed market value. I include single-family residences first sold by a developer in the 2000 to 2007 period in developments with an integrated lender. All specifications include month-of-sale fixed effects and county fixed effects. Controls for house characteristics (H), buyer characteristics (B), financing characteristics (F), and census tract demographics (T) are defined as in Table II. D1 includes developer fixed effects, D2 includes development fixed effects. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Integrated Lender	0.395*** (0.111)	0.445*** (0.101)	0.457*** (0.091)	0.367*** (0.088)	0.285*** (0.077)	0.173*** (0.046)
Controls		H	H, B, F	H, B, F	H, B, F	H, B, F
				T	T, D1	T, D2
R ²	0.811	0.828	0.832	0.881	0.892	0.936
$\bar{y}$	-6.346	-6.347	-6.347	-6.347	-6.347	-6.347
N	87,601	87,601	87,601	87,601	87,601	87,601

Table III: tax-assessment returns

Table IV
Relative Foreclosure Probability of Integrated Lender Mortgages

This table shows average marginal effects from probit regression (4). The dependent variable is whether a foreclosure was observed within three years of purchase. I include single-family residences sold by a developer over the 2000 to 2007 period in developments with an integrated lender. Each specification controls for month-of-sale fixed effects and county fixed effects. House characteristics (H), buyer characteristics (B), financing characteristics (F), and census tract demographics (T) are defined as in Table II. D1 includes developer fixed effects, D2 includes development fixed effects. Columns (5) and (6) restrict the sample to mortgages securitized in the year in which they were originated. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	SECURITIZED					
	(1)	(2)	(3)	(4)	(5)	(6)
Integrated Lender	-0.011*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.010*** (0.004)	-0.018*** (0.005)
Controls		H,B,F	H,B,F	H,B,F	H,B,F	H,B,F
			T,D1	T,D2	T,D1	T,D2
$\bar{y}$	0.017	0.018	0.018	0.021	0.019	0.028
N	71,714	70,397	68,354	59,319	12,322	8,638

Table IV: foreclosure probability

6.4 Tables V and VI: rejection evidence

Read:

- Table V shows that houses bought by borrowers rejected by integrated lenders perform worse and are more likely to enter foreclosure.
- Table VI shows that rejection by an integrated lender is predictive of low future collateral returns, but rejection by a nonintegrated lender is not.
- This is direct evidence that integrated lenders' private decisions contain information about collateral quality.

Economic message: The rejected-borrower analysis converts the integrated lender's screening decision into an observable signal. Integrated-lender rejections are informative; nonintegrated-lender rejections are not.

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Table V
Effect of Rejection by Integrated Lender

Columns (1) to (3) show results from regression (1), where the dependent variable is the annualized return of houses between two arms-length transactions. The controls are as in Table II. Columns (4) to (6) show average marginal effects from probit regression (4), where the dependent variable is whether a foreclosure was observed within three years of purchase. The controls are as in Table IV. I include single-family residences sold by a developer over the 2000 to 2007 period in developments with an integrated lender. The excluded category consists of houses financed by nonintegrated lenders, where the owner applied for and received a mortgage offer from the integrated lender but where this offer was dominated by an offer from a nonintegrated lender. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Return: Period A			Foreclosure		
	(1)	(2)	(3)	(4)	(5)	(6)
Integrated Lender	0.387*** (0.118)	0.409*** (0.120)	0.389*** (0.114)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.002)
Nonintegrated Lender + Rejection by IL	-0.450** (0.214)	-0.429** (0.211)	-0.368* (0.212)	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)
Nonintegrated Lender + No Application to IL		0.083 (0.154)	0.120 (0.142)		-0.002 (0.002)	-0.002 (0.002)
Controls	H,B,F, T,D1	H,B,F, T,D1	H,B,F, T,D1	H,B,F, T,D1	H,B,F, T,D1	H,B,F, T,D2
R^2	0.887	0.887	0.896			
$\bar{y}$	7.450	7.450	7.450	0.018	0.018	0.021
N	30,398	30,398	30,398	68,354	68,354	59,319

Table V: effect of rejection by integrated lender

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Table VI
Informativeness of Rejections

This table shows the effect of being rejected by a lender on the return over period A in Figure 1 in columns (1) and (2), and the probability of a foreclosure within three years in columns (3) and (4). I include single-family residences sold by a developer over the 2000 to 2007 period in developments with an integrated lender. In columns (1) and (3), I restrict the sample to houses where the buyers applied for a mortgage with the integrated lender; in columns (2) and (4), I restrict to houses where the buyers applied for a mortgage with a nonintegrated lender. The controls are as in column (4) of Table II for columns (1) and (2), and as in column (3) of Table IV for columns (3) and (4). Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Return: Period A		Foreclosure	
	(1)	(2)	(3)	(4)
Rejected by Integrated Lender	-0.376** (0.165)		0.007*** (0.001)	
Rejected by Nonintegrated Lender		0.018 (0.173)		0.003 (0.003)
Controls	✓	✓	✓	✓
Sample Applied to:	Integrated Lender	Nonintegrated Lender	Integrated Lender	Nonintegrated Lender
R^2	0.893	0.885		
$\bar{y}$	7.450	6.444	0.018	0.029
N	22,844	12,066	48,477	21,876

Table VI: informativeness of rejections

6.5 Tables VII and VIII: second owners and investment

Read:

- Table VII studies the second ownership period and finds continued performance differences even after the original borrower is gone.
- The foreclosure result also persists for second-owner outcomes.
- Table VIII uses 2008–2009 assessment changes and building permit data.
- There is no evidence that integrated-lender borrowers invest more in the house after purchase.

Economic message: The second-owner and permit results point away from borrower selection and differential maintenance, and toward collateral quality.

Table VII
Annualized Return (%): Second Ownership Period

This table shows results from a regression of the annualized return and foreclosure probability of houses during the ownership period of the second owner. I include all single-family residences initially sold by a developer over the 2000 to 2007 period in developments with an integrated lender. "Integrated Lender" is equal to one if the mortgage to the first owner was made by an integrated lender. Columns (1) and (2) measure the return from the second sale to the assessed value in January 2009, and restrict attention to houses resold prior to 2008. Columns (3) and (4) measure the return between the second and third sales. In columns (5) and (6), the dependent variable is an indicator for whether a foreclosure was observed within three years of purchase by the second owner. Control variables include fixed effects for month of resale in columns (1), (2), (5), and (6), and sales-quarter pairs in columns (3) and (4). All specifications include fixed effects for construction quarter and county as well as characteristics of the house (H) and census tract (T) as in Table II, and characteristics of the buyer (B2) and mortgage (F2). Buyer characteristics include whether the second buyer was an individual or a couple and whether the second buyer was Asian or Latino, in addition to characteristics of the first buyer (B) as in Table II. Financing characteristics include the loan-to-value ratio of the second mortgage in addition to details of the first mortgage (F) as in Table II. D1 includes developer fixed effects, D2 includes development fixed effects. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Return: Period C		2 nd to 3 rd Sale		Foreclosures	
	(1)	(2)	(3)	(4)	(5)	(6)
Integrated Lender	0.296*** (0.084)	0.178** (0.086)	0.463* (0.252)	0.320 (0.231)	-0.025*** (0.007)	-0.024*** (0.009)
Controls	✓	✓	✓	✓	✓	✓
Other Fixed Effects	D1	D2	D1	D2	D1	D2
R^2	0.902	0.947	0.881	0.891		
$\bar{y}$	-10.87	-10.87	3.323	3.323	0.061	0.069
N	18,339	18,339	5,390	5,390	9,537	8,358

Table VII: second ownership period

Table VIII
Annualized Return (%): 2008 to 2009 Assessment, Period D

The dependent variable in columns (1) to (3) is the return between the assessed market values in January 2008 and 2009, as given by regression (5), that in column (4) is a dummy variable for whether building size changed in 2008, and that in column (5) is a dummy variable for whether a building permit was filed in 2008. I include single-family residences first sold by a developer over the 2000 to 2007 period in developments with an integrated lender. In columns (1) to (4) I restrict the sample to properties in Maricopa, Pima, and Yuma Counties, for which I observe the 2008 assessed values. In column (5) I restrict the sample to properties in the city of Phoenix, for which I can observe building permits. All specifications control for the assessed house value in January 2008, in addition to other characteristics of the house (H), the buyer (B), the financing (F), and the census tract (T) as in Table II. D1 includes developer fixed effects, D2 development fixed effects. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Return: Period D			Δ Building Area (4)	Has Permit (5)
	(1)	(2)	(3)		
Integrated Lender	0.633*** (0.196)	0.334*** (0.127)	0.326*** (0.122)	-0.0006 (0.0010)	0.0002 (0.0027)
Δ Building Area			0.961 (0.818)		
Controls	D1	D2	D2	D2	D2
R^2	0.613	0.797	0.801	0.103	0.331
$\bar{y}$	-27.45	-27.45	-27.47	0.0042	0.0141
N	69,949	69,949	69,876	69,876	7,654

Table VIII: 2008–2009 assessment and permits

6.6 Tables IX and X: expansive soil and mortgage spread controls

Read:

- Table IX shows that the integrated-lender collateral advantage is larger for houses built on expansive soil, where construction quality is especially important.
- Table X controls for initial mortgage interest spreads among ARM and hybrid ARM loans.
- The integrated-lender collateral effect remains after controlling for mortgage spreads.

Economic message: Expansive soil identifies a setting where private construction-quality information should be particularly valuable. Interest-rate controls show that omitted public quality signals do not explain the main result.

Table IX
Annualized Return (%) by Soil Type

This table shows results from regression (6). ES_i is equal to one for houses built on hydrologic soil group D. Columns (1) and (2) correspond to Table II, columns (3) and (4) correspond to Table III, columns (5) and (6) correspond to Table VII, and columns (7) and (8) correspond to Table VIII. Control variables are included as indicated. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Period A		Period B		Period C		Period D	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Expansive Soil	-0.222 (0.509)	-0.768** (0.338)	-1.037* (0.598)	-0.896*** (0.187)	-1.843*** (0.406)	-0.812*** (0.284)	-3.352*** (1.272)	-0.827 (0.643)
Integrated Lender	0.383*** (0.119)	0.353*** (0.107)	0.224*** (0.060)	0.145*** (0.046)	0.213** (0.089)	0.168* (0.084)	0.493** (0.147)	0.116 (0.105)
Integrated Lender $\times$ Expansive Soil	0.564*** (0.196)	0.324 (0.227)	0.595*** (0.207)	0.293** (0.125)	0.778*** (0.255)	0.0551 (0.151)	1.274*** (0.448)	0.683** (0.307)
Controls	Table II	Table III	Table III	Table VII	Table VII	Table VIII	Table VIII	Table VIII
R^2	Col (4)	Col (5)	Col (6)	Col (5)	Col (4)	Col (1)	Col (2)	Col (2)
$\bar{y}$	0.887	0.896	0.891	0.906	0.902	0.947	0.854	0.831
N	7,450	7,450	6,347	-6.347	-10.87	-10.87	-28.09	-28.09
	30,398	30,398	87,601	87,601	18,339	18,339	69,949	26,869

Table IX: expansive soil heterogeneity

Table X
Test for Differential Riskiness: Control for Interest Rate

This table shows results when controlling for the initial interest rate, for the subset of ARMs and hybrid ARMs for which this information is available. The dependent variables in columns (1) to (4) are the annualized collateral return over periods A to D, respectively. The dependent variable in column (5) is the probability of observing a foreclosure within three years of the initial sale. The control variables are as indicated. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Period A	Period B	Period C	Period D	Foreclosure
	(1)	(2)	(3)	(4)	(5)
Integrated Lender	0.462* (0.238)	0.340*** (0.098)	0.471*** (0.159)	0.395** (0.170)	-0.007** (0.003)
Mortgage Spread	-0.450*** (0.099)	-0.167*** (0.027)	-0.086 (0.060)	-0.134* (0.078)	0.011*** (0.001)
Controls	Col (4)	Col (5)	Col (2)	Col (2)	Col (2)
	Table II	Table III	Table VII	Table VIII	Table IV
R^2	0.888	0.881	0.869	0.593	
$\bar{y}$	5.370	-8.340	-14.43	-27.59	0.038
N	8,002	23,864	3,439	6,885	16,012

Table X: interest-rate controls

6.7 Table XI, Figure 3, Table XII, and Figure 4: collateral quality and pricing responses

Read:

- Table XI shows that integrated-lender houses are less likely to be listed as damaged, “as-is,” or attractive to special buyers.
- Figure 3 shows that integrated-lender relative collateral returns vary by LTV group.

- Table XII shows that nonintegrated lenders charge higher rates when competing against integrated lenders.
- Figure 4 shows that this interest-rate premium rises strongly with the LTV ratio.

Economic message: The price response completes the adverse-selection story. Uninformed lenders appear to recognize that competition against informed lenders worsens the quality of the collateral they receive, especially when high LTV makes collateral risk more costly.

Table XI
Evidence for Property Damage

This table shows the average marginal effects from a probit regression explaining damage indicators in property listings. I analyze three indicators of property damage: whether the house is sold "as-is" (columns (1) and (2)), whether there were details of damage in the description (columns (3) and (4)), and whether the property was said to be attractive for special buyers (columns (5) and (6)). I include single-family residences first sold by a developer in the 2000 to 2007 period in developments with an integrated lender and that were listed for resale on Trulia.com between October 2005 and August 2010 (but at least one year after the first sale), and that include a textual description. All specifications include month of sale fixed effects, county fixed effects, and a control for the time between the initial sale and the listing. They also include house characteristics (H), buyer characteristics (B), financing characteristics (F), and census tract demographics (T) as in Table II. D1 includes developer fixed effects, D2 includes development fixed effects. Standard errors are clustered at the developer level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	"As-Is"		Damage Indicator		Special Buyer	
	(1)	(2)	(3)	(4)	(5)	(6)
Integrated Lender	-0.027*** (0.007)	-0.025*** (0.008)	-0.009** (0.005)	-0.010* (0.006)	-0.012** (0.005)	-0.016*** (0.006)
Controls	D1	D2	D1	D2	D1	D2
$\bar{y}$	0.138	0.143	0.055	0.063	0.046	0.055
N	11,299	10,742	10,908	9,379	10,759	8,808

Table XI: property damage evidence

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Table XII
Impact of Integrated Lender on Interest Rates of Other Lenders

This table shows results from regression (9). The dependent variable is the mortgage interest rate. I include single-family residences sold by a developer in Arizona over the 2000 to 2007 period that were financed by nonintegrated lenders. Each specification includes month $\times$ rate-type (adjustable or hybrid adjustable) fixed effects, county fixed effects, and lender fixed effects. Finance characteristics (F), house characteristics (H), buyer characteristics (B), and census tract demographics (T) are as in Table II. D1 includes developer fixed effects. Standard errors are clustered at the lender level and presented in parenthesis. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Has Integrated Lender	0.116** (0.055)	0.114** (0.054)	0.099** (0.045)	0.088** (0.043)	0.092** (0.042)	0.077* (0.042)
Expansive Soil			-0.000 (0.036)	0.023 (0.038)	0.001 (0.033)	-0.089* (0.048)
Has Integrated Lender $\times$ Expansive Soil						0.150*** (0.054)
Controls	F	F	F, H, B	F, H, B, T	F, H, B, T	F, H, B, T
R^2	0.555	0.583	0.591	0.592	0.597	0.592
$\bar{y}$	6.638	6.638	6.638	6.638	6.638	6.638
N	15,620	15,620	15,620	15,620	15,620	15,620

Table XII: interest-rate response

7 Short summary

- The paper studies asymmetric information about mortgage collateral values, not borrower characteristics.
- The setting is new-home purchases in Arizona developments where some developers have integrated mortgage lenders.
- Integrated lenders have private information about construction quality and collateral risk.
- Homes financed by integrated lenders have higher future returns and lower foreclosure rates.

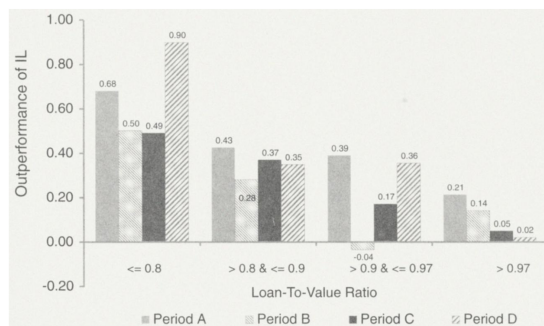


Figure 3: Relative collateral return of integrated lender by LTV ratio. This graph plots the κ_j coefficients from regression (8). I measure return over periods A, B, C, and D in Figure 1

Figure 3: collateral return by LTV

Asymmetric Information about Collateral Values

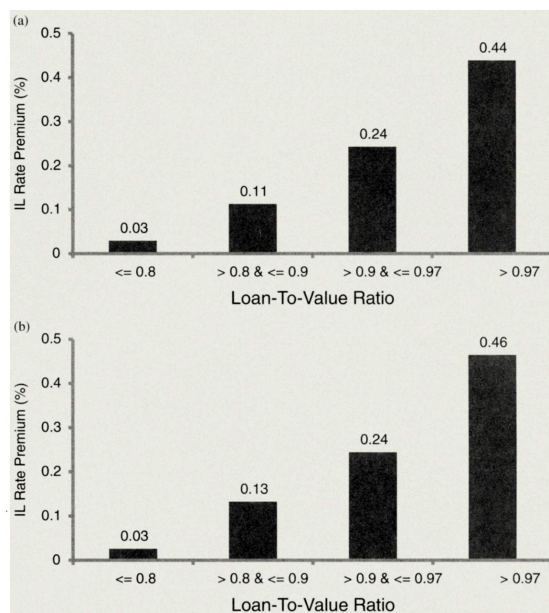


Figure 4: interest-rate premium by LTV

- Observable borrower characteristics do not explain the result; integrated-lender borrowers often look worse.
- Rejection by an integrated lender predicts worse collateral performance, while rejection by a nonintegrated lender does not.
- Second-owner returns, tax assessments, permit data, property listings, and expansive soil heterogeneity all support the collateral-information mechanism.
- Nonintegrated lenders charge higher interest rates when competing against integrated lenders, especially for high-LTV loans.
- The paper shows that informational asymmetry about collateral quality affects both allocations and prices in mortgage markets.

8 Conclusion

8.1 Core conclusion

Stroebel shows that integrated mortgage lenders possess economically important private information about collateral quality. They use this information to lend against better homes, leaving nonintegrated lenders exposed to adverse selection. This selection affects both ex post outcomes and ex ante mortgage pricing.

8.2 Economic intuition

Newly built homes can differ in unobservable construction quality. Developers and integrated lenders are better positioned to observe these differences than outside lenders. If the integrated lender declines to finance a home, outside lenders face a worse pool of collateral. In equilibrium, integrated lenders get better houses and nonintegrated lenders charge higher rates to compensate for adverse selection.

8.3 Takeaway

The paper broadens the empirical understanding of asymmetric information in credit markets. Credit risk depends not only on hidden borrower quality but also on hidden collateral quality. This matters for mortgage pricing, foreclosure risk, securitization, and policies that assume collateral values are publicly observable.